A Verification Note on a Three-Agent Weighted-Coverage Counterexample to Complete EFX

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Abstract

Status of the result. This note gives a complete negative solution, not merely partial progress, for the all-goods complete-EFX existence question under normalized monotone valuations. It exhibits three agents and eight goods with normalized monotone valuations for which no complete EFX allocation exists. In fact the valuations are weighted coverage valuations, hence monotone and submodular. The write-up is intended as a self-contained verification of the small cyclic weighted-coverage counterexample appearing in recent work of Mackenzie and Suzuki; no novelty of the construction itself is claimed.

1 The question and the theorem

Let M be a finite set of indivisible goods and let N be a finite set of agents. A valuation is a function

$$v_i : 2^M \to \mathbb{R}_{\geq 0}.$$

It is normalized if $v_i(\varnothing) = 0$, and monotone if $S \subseteq T$ implies $v_i(S) \leq v_i(T)$. A complete allocation is a tuple $X = (X_i)_{i \in N}$ of pairwise disjoint bundles with $\bigcup_i X_i = M$. It is EFX, envy-free up to any good, if

$$v_i(X_i) \geq v_i(X_j \setminus \{g\}) \qquad \text{for all } i, j \in N \text{ and all } g \in X_j. \tag{1}$$

Mahara's general-valuation question asks whether every finite instance with normalized monotone valuations admits a complete allocation satisfying (1). The theorem below answers this universal existence question negatively.

Theorem 1. *There is an instance with three agents and eight goods, with normalized monotone submodular valuations, for which no complete* EFX *allocation exists. The valuations can be chosen to be weighted coverage valuations. After rescaling, they also satisfy $v_i(M) = 1$ for every agent.*

Remark 1 (Attribution and scope). The construction verified here is the compact cyclic weighted-coverage counterexample of Mackenzie and Suzuki. It solves the unrestricted complete-allocation question for general monotone valuations by counterexample. It does not contradict positive complete-EFX results for special domains, such as additive valuations in certain low-agent settings, nor does it address approximate or partial-allocation variants.

2 A typed ordinal certificate

Let the goods be

$$M = A \cup B \cup C \cup \{x, y\}, \qquad |A| = |B| = |C| = 2,$$

with all displayed parts disjoint. We write type words such as $AABx$ to mean a bundle containing two goods of type A, one of type B, and the good x. Similarly, a support word such as ABx means that the represented types are A, B, x, regardless of multiplicity.

We first define a monotone ordinal rank function $r_0 : 2^M \to \{0, 1, \dots, 7\}$. Put $r_0(\varnothing) = 0$ and $r_0(\{g\}) = 1$ for each good g. The pair ranks are given by Table 1; diagonal entries for A, B, C refer to the pair of the two distinct goods of that type.

	A	B	C	x	y
A	1	2	2	4	6
B	2	1	5	1	3
C	2	5	1	1	3
x	4	1	1	-	1
y	6	3	3	1	-

Table 1: Pair ranks for r_0.

A triple is called exceptional if it is of type ABC or BCx. Equivalently, it contains one good from each of A, B, C, or it contains one B-good, one C-good, and x. For triples and larger sets define

$$r_0(S) = \begin{cases} 7, & |S| = 3 \text{ and } S \text{ is exceptional,} \\ \max_{\{g,h\} \subseteq S} r_0(\{g, h\}), & |S| = 3 \text{ and } S \text{ is not exceptional,} \\ \max_{T \subseteq S,\ |T|=3} r_0(T), & |S| \geq 4. \end{cases} \tag{2}$$

This rank is monotone: when a good is added, all previously present internal pairs or triples remain available in the maximum defining the new rank.

Choose a cyclic relabeling σ sending $A \mapsto B \mapsto C \mapsto A$ and fixing x, y. Concretely, after writing

$$A = \{a_1, a_2\}, \quad B = \{b_1, b_2\}, \quad C = \{c_1, c_2\},$$

one may take

$$\sigma = (a_1\ b_1\ c_1)(a_2\ b_2\ c_2).$$

Define

$$r_1(S) = r_0(\sigma(S)), \qquad r_2(S) = r_0(\sigma^2(S)).$$

Definition 1. An allocation $X = (X_0, X_1, X_2)$ is ordinal-EFX for (r_0, r_1, r_2) if

$$r_i(X_i) \geq r_i(X_j \setminus \{g\}) \qquad \text{for all } i, j \in \{0, 1, 2\} \text{ and all } g \in X_j.$$

A strict violation of this inequality will be called strong ordinal envy.

3 No ordinal-EFX allocation

We prove that the ordinal profile has no complete ordinal-EFX allocation. This is the combinatorial core of the counterexample.

Lemma 1 (First-pair restriction)**.** *Suppose that an allocation $X = (X_0, X_1, X_2)$ passes all ordinal-*EFX *inequalities involving agent* 0*, and suppose that*

$$|X_0| = 2, \qquad |X_1|, |X_2| \geq 2.$$

Then the type of X_0 is one of

$$Ax, Ay, BC, \qquad By, \qquad Cy.$$

Proof. By Table 1, the pair types of r_0-rank at least 3 are exactly

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

It is therefore enough to prove $r_0(X_0) \geq 3$.

Since eight goods are allocated and $|X_0| = 2$, the other two bundle sizes are either $3, 3$ or $2, 4$. First suppose they are $3, 3$. If $y \notin X_0$, let F be the triple containing y. Among the two goods of $F \setminus \{y\}$, at least one has type A, B, or C. Deleting the other good from F leaves a pair of type Ay, By, or Cy, all of rank at least 3. Agent 0's EFX inequalities force $r_0(X_0) \geq 3$.

If $y \in X_0$ and $r_0(X_0) < 3$, then necessarily $X_0 = xy$. The remaining goods are exactly $A \cup B \cup C$. No triple drawn from these six goods can have all three goods of the same type, because each type has only two goods. Thus every such triple contains a pair of type AB, AC, or BC, of rank at least 2. Since $r_0(xy) = 1$, agent 0 would strongly envy some deleted triple. This contradiction proves the claim in the $(3, 3)$ case.

Now suppose that one of X_1, X_2, say F, has size 4. Assume for contradiction that $r_0(X_0) \leq 2$. Then every pair contained in F must have rank at most 2: any such pair remains inside a suitable deletion $F \setminus \{g\}$, and agent 0's EFX test bounds every deletion by $r_0(X_0)$. Hence F avoids the five pair types

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

The only four-good type multisets with this property are $AABB$ and $AACC$. In either case, some deletion of F has rank 2, so $r_0(X_0) = 2$. If $F = AABB$, then X_0 is a pair selected from C, C, x, y, and no pair among these goods has rank 2. If $F = AACC$, the same argument applies with B, B, x, y. This contradiction gives $r_0(X_0) \geq 3$. □

Proposition 1. *There is no complete allocation of the eight goods that is ordinal-*EFX *for* (r_0, r_1, r_2).

Proof. The profile has cyclic symmetry. If $X = (X_0, X_1, X_2)$ is ordinal-EFX, then

$$X^\sigma = (\sigma(X_1), \sigma(X_2), \sigma(X_0))$$

is also ordinal-EFX, because $r_{i+1}(S) = r_i(\sigma(S))$ for indices modulo 3. Thus we may rotate the ordered bundle sizes cyclically.

First, no bundle can have size at most 1. If, after a cyclic rotation, $X_0 = \varnothing$, then some other bundle has at least two goods, and deleting one good leaves a nonempty bundle of rank at least $1 > 0 = r_0(X_0)$. If $|X_0| = 1$, then $r_0(X_0) = 1$, while one of the other bundles has size at least 4. Every four-good bundle has a deletion of rank at least 2: otherwise all its pairs would have rank at most 1, but a four-good set cannot be formed using only mutually rank-one pair types. Hence agent 0 again strongly envies after a deletion.

Therefore the only possible size multisets are $\{2, 2, 4\}$ and $\{2, 3, 3\}$. By cyclic symmetry, it remains to rule out the ordered patterns $(2, 2, 4)$ and $(2, 3, 3)$ with $|X_0| = 2$. Lemma 1 then restricts the type of X_0 to

$$Ax, \quad Ay, \quad BC, \quad By, \quad Cy.$$

The pattern $(2, 2, 4)$. Assume $|X_0| = |X_1| = 2$ and $|X_2| = 4$. Agent 0's inequalities imply that every deletion of X_2 has r_0-rank at most $r_0(X_0)$. Equivalently, X_2 contains no exceptional triple and no pair whose rank exceeds $r_0(X_0)$. Applying this filter to the five possible types of X_0 gives exactly the cases in Table 2. In every non-impossible case, the displayed inequality is a strict violation for agent 1, whose own bundle is X_1.

X_0	all possible $X_2 \mid X_1$	strict violation
Ax	impossible: no four-bundle can avoid both Ay and BC	–
Ay	$BBCC \mid Ax$	$r_1(Ax) = 1 < 2 = r_1(BCC)$
	$ACCx \mid BB$	$r_1(BB) = 1 < 4 = r_1(CCx)$
	$ABBx \mid CC$	$r_1(CC) = 1 < 5 = r_1(ABB)$
BC	$AACx \mid By$	$r_1(By) = 3 < 4 = r_1(ACx)$
	$AABx \mid Cy$	$r_1(Cy) = 6 < 7 = r_1(ABx)$
By	$AACC \mid Bx$	$r_1(Bx) = 1 < 2 = r_1(ACC)$
Cy	$AABB \mid Cx$	$r_1(Cx) = 4 < 5 = r_1(ABB)$

Table 2: All cases in the size pattern $(2, 2, 4)$. For instance, the witness BCC means that one B-good is deleted from $BBCC$.

Thus the pattern $(2, 2, 4)$ is impossible.

The pattern $(2, 3, 3)$**.** Assume $|X_0| = 2$ and $|X_1| = |X_2| = 3$. We again inspect the five possible types of X_0.

If $X_0 = Ax$, then the remaining types are A, B, B, C, C, y, and neither triple can contain a BC pair or an Ay pair. Therefore the split is either

$$ABB \mid CCy \qquad \text{or} \qquad ACC \mid BBy.$$

In the first split, the holder of ABB strongly envies the holder of CCy after deletion of a C-good, because $r_i(ABB) < r_i(Cy)$ for both $i = 1, 2$. In the second split, the holder of ACC strongly envies the holder of BBy after deletion of a B-good, because $r_i(ACC) < r_i(By)$ for both $i = 1, 2$.

If $X_0 = BC$, then the remaining types are A, A, B, C, x, y. Since a triple containing y cannot also contain an A-good, the split is one of

$$BCy \mid AAx, \qquad Bxy \mid AAC, \qquad Cxy \mid AAB.$$

In these three cases, respectively, the holder of the second displayed triple strongly envies the holder of the first after deletion of B, x, and x. The needed inequalities are

$$\begin{aligned}
(r_1(AAx), r_2(AAx)) &= (1, 1) < (6, 3) = (r_1(Cy), r_2(Cy)),\\
(r_1(AAC), r_2(AAC)) &= (2, 5) < (3, 6) = (r_1(By), r_2(By)),\\
(r_1(AAB), r_2(AAB)) &= (5, 2) < (6, 3) = (r_1(Cy), r_2(Cy)).
\end{aligned}$$

If $X_0 = Ay$, the remaining types are A, B, B, C, C, x. Let R be the triple containing the unique remaining A-good, and let S be the other triple. Suppose first that S is held by agent 1. Then $r_1(S) \le 4$. If R contains an AB pair, deletion of the third good leaves rank $r_1(AB) = 5$, so agent 1 strongly envies the holder of R. If R contains no AB pair, then R is ACC or ACx, and

$$r_1(BBx) < r_1(AC), \qquad r_1(BBC) < r_1(Cx).$$

Again agent 1 strongly envies the holder of R. The same argument, after cyclic relabeling, handles the case in which S is held by agent 2: if R contains AC, use $r_2(AC) = 5$; otherwise $R = ABB$ or ABx, and

$$r_2(CCx) < r_2(AB), \qquad r_2(BCC) < r_2(Bx).$$

Thus strong envy is unavoidable when $X_0 = Ay$.

If $X_0 = By$, no remaining triple can contain Ax or BC, and the split is forced as

$$CCx \mid AAB.$$

The holder of CCx strongly envies the holder of AAB after deletion of an A-good, since $r_i(CCx) < r_i(AB)$ for both $i = 1, 2$. Finally, if $X_0 = Cy$, the split is forced as

$$BBx \mid AAC,$$

and the holder of BBx strongly envies the holder of AAC after deletion of an A-good, since $r_i(BBx) < r_i(AC)$ for both $i = 1, 2$.

The two possible size patterns both lead to a strict violation. Hence no complete ordinal-EFX allocation exists. □

4 Realization by weighted coverage valuations

We now realize the ordinal obstruction by explicit cardinal valuations.

For $R \subseteq M$, define the coverage indicator

$$\chi_R(S) = \mathbf{1}\{S \cap R \neq \varnothing\}.$$

Define $u_0 : 2^M \to \mathbb{R}_{\geq 0}$ by

$$\begin{aligned} u_0(S) = \chi_B(S) &+ \chi_C(S) \\ &+ 3\chi_{A\cup\{x\}}(S) + 3\chi_{B\cup\{y\}}(S) + 3\chi_{C\cup\{y\}}(S) \\ &+ 8\chi_{A\cup B}(S) + 8\chi_{A\cup C}(S) \\ &+ 9\chi_{B\cup\{x,y\}}(S) + 9\chi_{C\cup\{x,y\}}(S) \\ &+ 2\chi_{A\cup B\cup\{y\}}(S) + 2\chi_{A\cup C\cup\{y\}}(S). \end{aligned}$$

Define

$$u_1(S) = u_0(\sigma(S)), \qquad u_2(S) = u_0(\sigma^2(S)).$$

Each u_i is a nonnegative linear combination of coverage indicators, so each u_i is a weighted coverage valuation.

Lemma 2. *The valuations u_0, u_1, u_2 are normalized, monotone, and submodular.*

Proof. Each coverage indicator χ_R satisfies $\chi_R(\varnothing) = 0$ and is monotone. It is also submodular: for $S \subseteq T$ and $g \notin T$, the marginal contribution

$$\chi_R(S \cup \{g\}) - \chi_R(S)$$

is 1 exactly when $g \in R$ and $S \cap R = \varnothing$, while the corresponding marginal at T is no larger. Nonnegative sums preserve normalization, monotonicity, and submodularity, and cyclic relabeling preserves these properties. □

The next lemma is the bridge from the ordinal certificate to the cardinal valuations.

Lemma 3 (Strict refinement)**.** *For every agent $i \in \{0, 1, 2\}$ and all bundles $S, T \subseteq M$,*

$$r_i(S) > r_i(T) \quad \Longrightarrow \quad u_i(S) > u_i(T).$$

Proof. It suffices to prove the assertion for agent 0, since the other two agents are obtained by cyclic relabeling. Both $r_0(S)$ and $u_0(S)$ depend only on the type support

$$\mathrm{supp}(S) \subseteq \{A, B, C, x, y\}.$$

Table 3 lists all 32 supports and their possible values. The value ranges are strictly separated by rank.

r_0	possible u_0-values	supports, with values where useful
0	0	$\emptyset$
1	21,23,28,31,35	$x(21)$; $A, B, C(23)$; $y(28)$; $xy(31)$; $Bx, Cx(35)$
2	36	AB, AC
3	37,40	$By, Cy(37)$; $Bxy, Cxy(40)$
4	41,45	$Ax(41)$; $ABx, ACx(45)$
5	46	BC, BCy
6	47,48	$Ay, Axy(47)$; $ABy, ACy, ABxy, ACxy(48)$
7	49	$ABC, ABCx, ABCy, ABCxy, BCx, BCxy$

Table 3: The support table for r_0 and u_0.

Thus

$$0 < 21 \leq 35 < 36 < 37 \leq 40 < 41 \leq 45 < 46 < 47 \leq 48 < 49.$$

Consequently, if $r_0(S) > r_0(T)$, then $u_0(S) > u_0(T)$. Relabeling gives the same implication for $i = 1, 2$. □

Proof of Theorem 1. Suppose, for contradiction, that there is a complete allocation $X = (X_0, X_1, X_2)$ that is EFX for the cardinal valuations (u_0, u_1, u_2). By Proposition 1, X is not ordinal-EFX. Hence for some agents i, j and some good $g \in X_j$,

$$r_i(X_j \setminus \{g\}) > r_i(X_i).$$

By Lemma 3,

$$u_i(X_j \setminus \{g\}) > u_i(X_i),$$

contradicting the EFX inequality for the cardinal valuation u_i. Therefore no complete EFX allocation exists for (u_0, u_1, u_2).

Finally, $u_i(M) = 49$ for all three agents. Setting $v_i = u_i/49$ preserves every EFX comparison and gives normalized monotone submodular valuations with $v_i(\varnothing) = 0$ and $v_i(M) = 1$. This proves the theorem. □

5 Conclusion

The result is a full counterexample to the universal complete-EFX existence statement for general normalized monotone valuations. It is stronger than required for that problem because the valuations are weighted coverage functions, and hence are monotone submodular. Thus the original all-goods existence question has a negative answer; any positive theorem must either restrict the valuation domain further or relax the allocation/fairness requirement.

A A finite verifier

The following script is not needed for the proof above, but it gives a direct reproducibility check. It enumerates all $3^8 = 6561$ complete allocations, checks the ordinal and cardinal EFX conditions, verifies strict refinement, and verifies submodularity of the cardinal valuations.

```
from itertools import combinations, product
from collections import Counter

M = set(range(8))
A = {0, 3}; B = {1, 4}; C = {2, 5}; x = {6}; y = {7}
sigma = {0: 1, 1: 2, 2: 0, 3: 4, 4: 5, 5: 3, 6: 6, 7: 7}

def sig(S, k=1):
    S = set(S)
    for _ in range(k):
        S = {sigma[g] for g in S}
    return S

def typ(g):
    if g in A: return "A"
    if g in B: return "B"
    if g in C: return "C"
    if g in x: return "x"
    return "y"

pair_rank = {
    ("A", "A"): 1, ("B", "B"): 1, ("C", "C"): 1,
    ("A", "B"): 2, ("A", "C"): 2, ("A", "x"): 4,
    ("A", "y"): 6, ("B", "C"): 5, ("B", "x"): 1,
    ("B", "y"): 3, ("C", "x"): 1, ("C", "y"): 3,
    ("x", "y"): 1,
}

def pr(a, b):
    return pair_rank.get((a, b), pair_rank.get((b, a)))

def r0(S):
    S = set(S)
    if not S:
        return 0
    if len(S) == 1:
        return 1
    if len(S) == 2:
        a, b = [typ(g) for g in S]
        return pr(a, b)
    if len(S) == 3:
        c = Counter(typ(g) for g in S)
        if c == Counter({"A": 1, "B": 1, "C": 1}):
            return 7
        if c == Counter({"B": 1, "C": 1, "x": 1}):
            return 7
        return max(r0(P) for P in combinations(S, 2))
    return max(r0(T) for T in combinations(S, 3))
```

```
def r(i, S):
    return r0(sig(S, i))

terms = [
    (1, B), (1, C),
    (3, A | x), (3, B | y), (3, C | y),
    (8, A | B), (8, A | C),
    (9, B | x | y), (9, C | x | y),
    (2, A | B | y), (2, A | C | y),
]

def u0(S):
    S = set(S)
    return sum(w for (w, R) in terms if S & R)

def u(i, S):
    return u0(sig(S, i))

def is_ordinal_efx(allocation):
    for i in range(3):
        own = r(i, allocation[i])
        for j in range(3):
            for g in allocation[j]:
                if own < r(i, allocation[j] - {g}):
                    return False
    return True

def is_cardinal_efx(allocation):
    for i in range(3):
        own = u(i, allocation[i])
        for j in range(3):
            for g in allocation[j]:
                if own < u(i, allocation[j] - {g}):
                    return False
    return True

allocations = []
for assignment in product(range(3), repeat=8):
    X = [set(), set(), set()]
    for good, agent in enumerate(assignment):
        X[agent].add(good)
    allocations.append(X)

assert sum(is_ordinal_efx(X) for X in allocations) == 0
assert sum(is_cardinal_efx(X) for X in allocations) == 0

# Precompute values for the remaining finite checks.
subsets = [frozenset(T) for k in range(9) for T in combinations(M, k)]
rval = {(i, S): r(i, S) for i in range(3) for S in subsets}
uval = {(i, S): u(i, S) for i in range(3) for S in subsets}
for i in range(3):
    for S in subsets:
        for T in subsets:
```

```
            assert not (rval[i, S] > rval[i, T] and uval[i, S] <= uval[i, T])
            assert uval[i, S] + uval[i, T] >= uval[i, S | T] + uval[i, S & T]

print("all checks passed")
```

References

[1] Ryoga Mahara. Extension of additive valuations to general valuations on the existence of EFX. *Mathematics of Operations Research*, 49(2):1263–1277, 2024. DOI: 10.1287/moor.2022.0044.

[2] Hannaneh Akrami, Alexander Mayorov, Kurt Mehlhorn, Shreyas Srinivas, and Christoph Weidenbach. A counterexample to EFX; $n \geq 3$ agents, $m \geq n + 5$ items, monotone valuations; via SAT-solving. arXiv:2604.18216, 2026.

[3] Simon Mackenzie and Mashbat Suzuki. Counterexamples to EFX for submodular and subadditive valuations. arXiv:2605.06451, 2026.